The Quiet Half of the Blast Radius: LLMs Catch the Blackhole but Miss the Silent Reroute in Network Configuration Changes

Technical Report v1

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Abstract

Large language models are beginning to write network configuration changes, and the safety question has so far been posed as a binary: *is the change safe or not?* We argue the operational question is graded and typed: *exactly which BGP sessions, prefixes, paths, and forwarding behaviors does this change damage?* We introduce *blast-radius prediction*: given device configurations and a proposed diff, a model must enumerate the typed set of damaged elements across nine scored damage dimensions—from BGP sessions, prefixes, and reachability to waypoint and redundancy loss, forwarding anomalies, resource exhaustion, control-plane withdrawals, and intent violations. Ground truth is computed by the Batfish network verifier on a digest-pinned, determinism-gated snapshot pair, and predictions are scored by deterministic, semantics-aware set comparison—no LLM judge. We evaluate six frontier and mid-tier models from three vendors on NetBrace, a benchmark of 11 incident-grounded scenarios and 48 generator-produced instances with randomized identifiers: 810 graded samples (N=5 per instance). The measured capability split is sharp and runs along an operationally meaningful line. Models read the *loud* half of the blast radius well: frontier models identify both grounded blackholes—the explicit leak-discard and the second-order partial-ECMP one—in every sample (10/10 each) and read first-order prefix loss at F1 0.73–0.83. The *quiet* half—damage that removes no nameable object and leaves reachability intact—goes largely unseen: a leak-induced silent reroute that pushes a victim's traffic off its mandated inspection waypoint, while every route still resolves and every session stays up, scores zero on both the waypoint and intent dimensions on 96 of 120 samples across the four models run on it—including both Anthropic frontier models (Opus 4.8 misses both on 21 of 30, despite reading the loud half perfectly). The damage models *do* catch shares a signature: it removes or blackholes something they can name—a withdrawn route, a vanished management prefix, a blackholed destination, which the stronger models catch at F1 1.00 where each is the ground truth (the two cheapest, 0.60–0.80). No model ever raises a false session-loss alarm (0/810). The most expensive real outages of the last decade—leak-induced silent reroutes among them—live in the quiet half. We additionally show that naive exact-string set grading would have reported a uniform 0.00 on forwarding anomalies for every model—a granularity artifact, not a capability fact—and release both graders with a frozen semantic-matching specification. The benchmark, oracle pipeline, and run artifacts will be released.

1 Introduction

Network operators are rapidly adopting LLM-based assistants that propose, review, and increasingly *apply* configuration changes. The emerging evaluation literature measures whether these

systems can *fix* things: configuration synthesis from intent [6], end-to-end repair of injected misconfigurations [2, 1], multi-turn configuration tasks [4], and incident troubleshooting [5]. Where safety appears, it is binary or aggregate: NetArena scores rule-adherence of agent actions in an emulator [3]; the agentic-repair study of Asadli et al. [1] reports a regression score—the fraction of the checked specification set that an *applied* repair newly breaks—computed *by the verifier* after the fact.

Production change review asks a different question. Before a change ships, a reviewer is expected to articulate its *blast radius*: which BGP sessions will drop, which prefixes will disappear from which routing tables, which flows will lose their path, which traffic will silently blackhole, and which device will exhaust its FIB. The damage is *typed* and *itemized*, and the most expensive failures in operational history are *quiet*: the configuration change looks innocuous, every BGP session stays established, often every destination even stays reachable—and the damage emerges only from the interaction of routing policy with topology. A leaked more-specific route attracts traffic onto a path that cannot deliver it—congesting it, as in the 2019 route-leak incidents, or discarding it outright, as we model here for determinism. A "cleanup" filter's implicit deny silently withdraws a customer prefix while the session stays up. A LOCAL_PREF adjustment quietly steers traffic around the inspection node it is required to transit.

This paper asks whether LLMs can see that damage *before* a change is applied, from the same evidence a human reviewer has: the device configurations and the diff. We make three contributions.

1. Task: typed blast-radius prediction. Given pre-change configurations and a proposed diff, the model must output the affected *set*—typed elements across nine scored damage dimensions (§2)—and is scored with per-dimension precision, recall, and F1 against verifier-computed ground truth. Correctly predicting *no* damage is a first-class outcome (several instances have empty ground truth by construction), and the cliché failure mode—hallucinating a BGP session loss—is tracked as an explicit false-session-alarm metric.

2. Benchmark and grading methodology: NetBrace. A two-split question base (§3): 11 hand-authored scenarios grounded in documented outage classes (route-reflector filter withdrawals, more-specific leaks, Clos summarization blackholes, FIB-overflow de-aggregation, management-plane lockout), plus 48 instances stamped by four parameterized generators that re-randomize ASNs, addressing, and names per seed as a contamination control. Ground truth is computed by Batfish [11] on a digest-pinned engine; every instance passes a $\geq 10\times$ byte-identical determinism gate before any paid model call. Grading is deterministic set arithmetic with *semantic canonicalization* (§2)—no LLM judge anywhere in the loop [14]. We show this canonicalization is not optional: naive exact-string matching against probe-granularity ground truth fabricates a uniform 0.00 on the forwarding-anomaly dimension for every model (§2, Appendix A)—a measurement pitfall we expect generalizes to other verifier-graded benchmarks.

3. Measurement: the quiet half of the blast radius is the blind spot. Across six models from three vendors (GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash, Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.4-mini; N=5 samples per instance; 810 graded samples), capability splits along the loud/quiet line, not the first-order/second-order line we expected. Loud damage is read well: every frontier model identifies both grounded blackholes—including the second-order, partial-ECMP one—on every sample, first-order prefix loss is read at F1 0.73–0.83, and damage that *removes* a nameable object—a withdrawn route, a vanished management prefix, a dropped ECMP path—is caught at F1 1.00 (stronger models; 0.60–0.80 for the cheapest) where it is the ground truth. Quiet

damage—where reachability survives and nothing the diff directly changes looks broken—is largely missed: traffic rerouted past a mandated waypoint by a leak *elsewhere*, with every route still resolving, scores zero on both the waypoint and intent dimensions on 96/120 samples across the four models run on it, both Anthropic frontier models included (Opus 4.8 misses both on 21/30 despite F1 1.00 on the loud half)—as does the traffic-pinning intent a surviving covering aggregate hides even as the pinned specifics are dropped (§5). The split is not first- vs. second-order (the Clos blackhole is second-order and caught); it is whether the damage leaves a trace. The errors are one-sided: the false-session-alarm rate is 0.00 across all 810 samples. Model-free floors (empty, naive, syntactic-diff, and a per-class memorizer) score $\leq$0.02 grounded and $\leq$0.08 templated, confirming the benchmark measures reasoning headroom rather than template recall, and the cross-vendor frontier lands within noise of itself, ruling out grader circularity.

We argue this split—competent at damage that announces itself in the routing table, blind to damage that hides behind preserved reachability—motivates a concrete *candidate graduation test* on the autonomy ladder for network agents (§6): before a system self-approves changes, it should first demonstrate it can enumerate the blast radius of its own proposals, *including the quiet half*—a bar no evaluated model currently meets.

2 Blast-Radius Prediction

Setting. A task instance is a triple: a set of device configurations C (one file per router, FRR dialect in this release), a proposed change expressed as a unified diff Δ over one or more devices, and a natural-language statement of operator intent. The model receives all three plus an output contract, in a single turn, with no tools; it must return a JSON object enumerating the damaged elements. The no-tools condition is deliberate: it isolates the model's *own* causal model of the network—the capability a pre-deployment reviewer exercises— from its ability to drive external tools, and it reflects the common deployment where an assistant reviews changes without a verifier in the loop. (A tools-augmented contrast is explicitly future work; §6.)

Damage dimensions. Damage is typed. Each element type defines a canonical string key (e.g., `bgp|vrf=default|node=r1|peer_ip=10.0.0.4`), making ground truth and predictions comparable as sets. Ten dimensions are declared; D6 is deferred in this release, so **nine are scored**, and every result in this report is over those nine:

- **D1 sessions**: BGP sessions established before, not after.
- **D2 prefixes**: watched prefixes present in a device's main RIB before, absent after.
- **D3 reachability**: flows delivered before, undeliverable after.
- **D4 waypoints**: flows that must transit a waypoint (e.g., an inspection node) and no longer do.
- **D5 redundancy**: (node, prefix) pairs whose installed equal-cost path count drops.
- **D6 packet filters**: permit↔deny transitions (*deferred* in this release: Batfish does not model FRR access-lists as interface filters; a vendor dialect is in progress).
- **D7 forwarding anomalies**: flows newly blackholed, null-routed, or looping while a route for them persists.
- **D8 scale**: post-change installed-route counts crossing a declared per-device resource budget (FIB/RIB).

Two auxiliary control-plane surfaces are scored alongside: **D9** BGP-RIB best-path withdrawals (control-plane loss masked from the FIB by, e.g., a static backup) and **D10** violations of declared intent invariants (reachability, waypoint, isolation, redundancy-floor, and traffic-pinning assertions evaluated oracle-side and never shown to the model).

Oracle. Ground truth is computed by Batfish [11] on a *snapshot pair*: the pre-change configurations and the post-change configurations produced by strict application of Δ. Sessions, RIBs, BGP-RIBs, route counts, and flow traces are queried in both worlds and diffed per dimension. Three properties make the oracle trustworthy as an answer key: (i) the engine is pinned by container digest, so ground truth is tied to exact bits; (ii) every instance passes a determinism gate—the full computation is repeated $\geq 10\times$ on fresh networks and must produce byte-identical key sets; (iii) scenario classes were spot-validated against live FRR containers (sessions, RIBs, traceroutes) before being trusted.

Grading. For each dimension we compute precision, recall, and F1 over element sets, with the empty-set convention: if truth and prediction are both empty the dimension scores 1.0 (correctly predicting *no* damage); a non-empty prediction against empty truth scores 0 (false alarm); an empty prediction against non-empty truth scores 0 (miss). The instance score is macro-F1 over *active* dimensions (non-empty in truth or prediction), so models cannot farm score by emitting empty lists for irrelevant dimensions; per-dimension means follow the same active convention.

Element matching is *semantics-aware*, per a specification frozen before any per-model number was computed (released with the artifacts). The oracle expresses flow-scoped truth (reachability, waypoints, forwarding anomalies, some intents) as probe *host routes* (/32s), but the output contract asks models for destination *prefixes*—the granularity the configurations actually expose. We therefore canonicalize each probe to its most-specific config-visible covering prefix and credit a prediction whose prefix lies in the band [probe, canonical cover]; all non-prefix fields (node, source, waypoint, VRF) still match exactly, and matching is deterministic one-to-one bipartite. The band is two-sided and not gameable: a prediction broader than the canonical cover earns nothing (no credit for `0.0.0.0/0`; an over-broad answer claims demonstrably-undamaged traffic is damaged and is charged as a false positive), and a narrower prediction earns credit only if it covers the probed host—which requires correctly reasoning about *which* traffic the change harms. For forwarding anomalies, `blackhole` and `null_routed` form one equivalence class (the contract's own gloss conflates them); `loop` stays distinct. The canonicalization is provably the identity on dimensions whose truth is already config-visible (D1, D2, D8, D9), and it leaves every model-free floor at chance: the largest floor movement under it is the syntactic-diff floor's $0.007 \rightarrow 0.023$ (templated), with all floors remaining ≤ 0.02 grounded and ≤ 0.08 templated (§3.3).

Two flags are tracked independently: *false session alarm* (predicting D1 loss when truth has none—the cliché reflex) and *invalid output* (no parseable JSON; reported separately so formatting failure is never conflated with reasoning failure). Parsing is tolerant (last JSON object in the response; missing keys are empty lists; malformed elements are dropped).

Why naive set grading fails here. Exact-string matching of prediction keys against probe-granularity truth keys is *unwinnable by construction*: a model that names the correct node, the correct anomaly kind, and the exact prefix the configuration exposes (e.g., the leaked /25) still scores 0 against a truth key carrying an undisclosed probe /32 inside that prefix. Under exact-string grading, the forwarding-anomaly F1 (active convention) is exactly 0.00 for every model—no forwarding-anomaly-truth sample earns any credit, even when the model names the leaked

Scenario	Mechanism class	Damage (truth)	Trap
ipc3	RR cleanup-filter implicit deny	D2, D3 (D1 = $\emptyset$)	✓
ipc3_static	same + static backup masks FIB	D9 only	✓
ipc_fanout	filter on whole peer group	D2×3, D3×3	✓
ipc_aggregate	/25s dropped, /24 aggregate survives	D2, D10	✓
leak_ms	more-specific leak → LPM blackhole	D3, D7	✓
clos_summary	spine summarization, partial ECMP blackhole	D2, D7	✓
scale_deagg	de-aggregation overflows FIB budget	D2, D8	✓
mgmt_lockout	mgmt prefixes omitted → NOC lockout	D3, D10	✓
lp_waypoint	LOCAL_PREF shift bypasses inspection	D4	✓
waypoint4	route shift bypasses waypoint (static lab)	D4	
ecmp4	one of two equal-cost paths removed (static lab)	D5	

Table 1: The grounded split. "Trap": the naive answer (a BGP session drops) is wrong; D1 ground truth is empty while damage occurs elsewhere. Every row's damage signature is verified against the committed oracle output. *waypoint4*/*ecmp4* are BGP-free static-routed anchors.

/25 against the undisclosed /32 probe—while the raw outputs show frontier models naming the blackholing node and the affected prefix on every such sample. We report the exact-string numbers in Appendix A as a conservative lower bound and a cautionary result for verifier-graded benchmark design: when an oracle's ground truth and a model's output contract live at different granularities, string-set scoring measures the mismatch, not the model.

3 The NetBrace Benchmark

3.1 Grounded split: 11 incident-anchored scenarios

Each hand-authored scenario instantiates a documented outage class on a small (4–6 node) topology whose configurations are verified against live FRR and whose ground truth passes the determinism gate. Table 1 summarizes them. Nine of eleven are *traps*: the change looks routine, every BGP session stays established (D1 truth is empty), and the real damage lives in D2–D10; the remaining two (*waypoint4*, *ecmp4*) are static-routed sanity anchors with no BGP at all, so their empty D1 is structural rather than adversarial. Examples: an iBGP route-map "cleanup" on a route reflector whose implicit deny silently withdraws a customer prefix (sessions: all established); a leaked more-specific /25 that attracts a victim /24's traffic into a discard (neither session nor prefix counts change); a Clos spine summarization that blackholes exactly one spine's share of ECMP traffic (the blackhole arises from a fixed wiring asymmetry that the summary masks—an instance of the summarization hazard documented in RFC 7938 [12]; the dynamic link-failure variant is future work); a de-aggregation event that overflows an edge FIB budget while reachability is fully preserved; and a management-prefix omission that locks the NOC out of every device (the scenario models in-band management reachability, the common and dangerous case).

3.2 Templated split: 48 generator instances

Four parameterized generators convert verified mechanisms into structurally distinct instances: *silent withdrawal* (fan-out $k \in \{2, 3, 4\}$; one-vs-all peers filtered; in/outbound; plain/static/aggregate prefix backing; implicit-deny vs explicit-deny vs off-by-one prefix-list boundaries), *more-specific leak* (blackhole vs *reroute* variants—in the latter, reachability survives while traffic silently shifts; leaked-scope variants), *attribute path-shift* (LOCAL_PREF and prepend knobs, including no-op variants

whose ground truth is *empty* by construction—calibration traps), and *de-aggregation overflow* (clearly-over, marginally-over, and under-budget variants). Every instance re-randomizes ASNs (random private 4-byte), addressing, node names, and the customer prefix per seed, so instances cannot be answered by recall of public configurations; topology identifiers shown to the model are content-free hashes. The split serves as the statistical backbone (real N per dimension) and the contamination control: a per-class memorizer floor (§3.3) directly tests whether template recognition suffices.

3.3 Model-free floors

Every sweep computes six deterministic baselines on the same instances: *empty* (predict nothing), *naive-session-loss* (predict exactly the cliché: one session drops at the changed device), *naive-count* (a counting heuristic over watched prefixes), *syntactic-diff* (regex over the diff text, zero routing knowledge), *modal-by-class* (replay the per-class median ground truth—a perfect memorizer of the generator's templates), and *oracle-ceiling* (echo the truth; a pipeline sanity check that must score exactly 1.0). Each floor operationalizes a specific skeptical hypothesis about what the benchmark "really" measures. All floors are evaluated under both graders; the semantic canonicalization leaves every floor at chance (§5).

4 Experimental Setup

Models. Six models from three vendors, pinned IDs as served in June 2026: GPT-5.5 and GPT-5.4-mini (OpenAI), Claude Opus 4.8, Sonnet 4.6, and Haiku 4.5 (Anthropic), and Gemini 3.5 Flash (Google). GPT-5.5 and GPT-5.4-mini run with reasoning enabled and accept no sampling parameters; Opus 4.8 has removed them; all three are sent no temperature. Haiku 4.5 and Sonnet 4.6 sample at temperature 0.7; Gemini 3.5 Flash is sent no temperature (API default). Gemini 3.5 Flash is Google's fast tier (the 3.5 Pro tier was not generally available on the API at sweep time); its scores land within noise of Opus 4.8 and GPT-5.5, so we group it with the frontier set. Single-shot prompting under one frozen prompt contract (frozen, with a leak-lint pass, before any paid call; the contract requires one JSON object with one key per dimension).

Protocol. N=5 samples per (instance, model). All six models ran the grounded split ($6 \times 11 \times 5 = 330$ samples); the templated split was run on Haiku 4.5 and GPT-5.4-mini ($2 \times 48 \times 5 = 480$ samples) under the study's fixed per-provider cost cap, for **810 graded samples** in total—the full-fleet templated run is gated on the in-progress human-expert validation study. To test the headline silent-reroute finding beyond the two templated-split models, the six-instance reroute class was additionally run on Opus 4.8 and Sonnet 4.6 ($2 \times 6 \times 5 = 60$ samples; GPT-5.5 and Gemini were not available for this extension), reported in F3. Both the prompt contract and the floor definitions were frozen before any paid call. The semantic-grader matching rules (§2) were motivated by the exact-string results and a raw-output audit (which showed models naming the node, kind, and config-visible prefix that the string grader rejected), were specified from the contract/oracle granularity gap rather than tuned against model scores, and were frozen before any per-model number was computed under them; the floor-invariance and config-visible-identity properties (F5) hold regardless of that timeline. Aggregation is two-stage: per-instance mean over samples, then mean ± population std across instances (with SEM = std/$\sqrt{n}$ used for significance)—never pooling samples as i.i.d. Per-dimension cells backed by a single instance are reported as point estimates, never as mean±std. We fix N=5 rather than higher: a variance decomposition shows the

grounded macro-F1 standard error is dominated by between-instance spread, not within-instance sample noise (within-instance std ≈0.02–0.03 for frontier models versus 0.06–0.17 across instances), so it changes by <0.001 for any $N > 5$; statistical power is therefore invested in instance count, not samples. Headline damage-detection macros exclude the empty-truth calibration instances (an empty predictor passes those for free); those instances are scored and reported separately via the false-alarm channel. Total model spend for the study was approximately $32, including a $3 re-run of the Sonnet grounded split and a $3 four-model reroute extension, both at the higher token cap (Disclosure ii; §5). The oracle container digest is pinned (`09817554db90…`) and recorded with every run manifest.

Disclosures. (i) Gemini's API returned transient 503 errors on a fraction of calls; its clean N=5 is pooled from three post-freeze runs under the identical frozen prompt and the same pinned model ID (draws are i.i.d. samples of one condition; only Gemini is pooled, and only for provider outages; partial and pooled estimates agree to within 0.01 macro-F1). (ii) The initial sweep capped Anthropic-path output at 4,000 tokens; Sonnet 4.6—the most verbose reasoner in the fleet—hard-truncated on 6/55 grounded samples (all emitting exactly 4,000 tokens, with the correct analysis, including the *clos_summary* blackhole, visible in the cut-off prose). We re-ran Sonnet's grounded split at a 12,000-token cap (parity with the other providers' budgets) under the identical frozen prompt; the re-run yielded zero invalid samples and is the source of Sonnet's reported grounded numbers. The oracle ground truth is byte-identical between the two runs (confirmed by the determinism gate), so only Sonnet's predictions differ; no other model approached the cap. (iii) One registry scenario (a static-route blackhole) was excluded from the paid sweep by a pre-registration audit that judged its topology contrived; it is an explicitly induced discard of the kind every frontier model detects reliably, so including it would have *raised* model scores on D7.

5 Results

F1: The benchmark discriminates. Frontier models score 0.48–0.51 damage-detection macro-F1 on the grounded split; the lower-tier models 0.27–0.42; all model-free floors ≤0.02 on that split (Table 2). On the templated split's damage instances, Haiku 4.5 and GPT-5.4-mini score 0.30±0.18 and 0.21±0.20 while the modal-by-class memorizer scores **0.00** and the best other floor 0.075 (Figure 3): the task is neither impossible nor saturated, and performance is not explained by template recall.

F2: Models do not cry wolf on sessions. The false-session-alarm rate is 0.00 for every model on every split it was evaluated on—all 810 samples. The widely-assumed failure mode—hallucinating session losses on any BGP-adjacent change—did not occur once, while the metric itself is demonstrably sensitive (the strawman floor trips it at 1.00). Models do, however, sometimes *double-type* real damage: on silent-withdrawal instances, models often report the dropped traffic both as reachability loss (correct—the frontier scores 1.0 there on the very same samples) and additionally as a forwarding anomaly—a typing the contract's own gloss ("blackhole = no route") invites but the oracle reserves for discarded-while-routed flows. We charge these as false positives—and the same taxonomy-boundary over-prediction extends to D9 and D10, where each model labels a consequence dimension on roughly six and eight of the eleven grounded instances the oracle reserves for a neighbor, depressing those active-convention dimensional means (F3, Appendix A) well below the recall the models actually achieve where the dimension is the truth. We flag the consequence-dimension boundaries for a sharpened v1.1 contract. On the empty-truth calibration

Model	macro-F1 (damage)	blackhole (D7†)	withdrawal (D9‡)	lockout (D10‡)	false sess. alarm	invalid
Claude Opus 4.8	0.51±0.17	**1.00**	**1.00**	**1.00**	0.00	0.00
Gemini 3.5 Flash	0.49±0.18	**1.00**	**1.00**	0.20	0.00	0.00
GPT-5.5	0.48±0.17	**1.00**	**1.00**	**1.00**	0.00	0.00
Claude Sonnet 4.6	0.42±0.06	**0.97**	**1.00**	**1.00**	0.00	0.00
Claude Haiku 4.5	0.39±0.10	0.57	0.80	0.76	0.00	0.00
GPT-5.4-mini	0.27±0.21	0.17	0.60	0.00	0.00	0.00
floors (best of 5)	≤0.02	0.00	0.00	0.00	1.00*	—

Table 2: Grounded split (11 scenarios, N=5, semantic grader). All three columns are *F1 on the instance(s) where the dimension is the ground truth* (recall is at least as high)—the loud half, which models read well. †Mean D7 F1 over the two anomaly-truth instances (*leak_ms*, *clos_summary*); four of six read both blackholes at ≥0.97, the two cheapest miss the implicit Clos one. ‡F1 on the single grounded instance where the dimension is the truth: the masked control-plane withdrawal (*ipc3_static*, D9) and the management-plane lockout (*mgmt_lockout*, D10). The *quiet* half (the silent reroute, missed) is templated and reported in §5 (F3); the low active-convention D7/D9/D10 *dimensional* means (Figure 1, F3) are precision-dragged by over-prediction, not recall failures (F3). *The naive-session-loss strawman trips the false-session-alarm metric at 1.00, demonstrating the metric detects the reflex; every real model passes it.

instances, the residual false alarms are sparse redundancy and intent over-predictions by the two models evaluated on that split; no model invents sessions, prefixes, or blackholes out of nothing.

F3: The blind spot is the quiet half, not the second order. We expected models to read first-order table effects and miss second-order consequences. The data draw the line elsewhere. The *loud* half is read well: every frontier model identifies both grounded blackholes on every sample (10/10 each on *leak_ms* and *clos_summary*—the latter a genuinely second-order, partial-ECMP consequence of a summarization change), Sonnet 4.6 nearly matches them (D7 0.97), and prefix-loss (D2) F1 is 0.73–0.83 across the frontier and Sonnet (0.77); even the two cheapest models catch the explicit leak blackhole (Haiku 1.00 grounded, 0.98 on the templated leak-blackhole class), though the two cheapest miss the implicit, topology-emergent one (*clos_summary*: Haiku 0.13, GPT-5.4-mini 0.13) that the frontier and Sonnet catch. Read where each *consequence* dimension is the ground truth, the picture is the same: the masked control-plane withdrawal (*ipc3_static*, D9) is caught at F1 1.00 by four models and 0.80/0.60 by the two cheapest; the management-plane lockout (*mgmt_lockout*, D10) at 1.00 by Opus, GPT-5.5, and Sonnet (Haiku 0.76); and the explicit LOCAL_PREF waypoint bypass (*lp_waypoint*, D4) at 1.00 by all six (Figure 2).

The *quiet* half is the damage that survives this object-level reading: reachability holds, and the harm is a path or policy the diff never spells out. Its sharpest case is the leak-induced silent reroute: on the templated reroute variants, where a more-specific leak *elsewhere* pulls a victim's traffic past its mandated waypoint while the route still resolves and every destination stays reachable, the two templated-split models miss it almost entirely (57/60 samples score 0 on both the waypoint and intent dimensions). We additionally ran the reroute class on the two Anthropic frontier models, Opus 4.8 and Sonnet 4.6: they do not escape it. Both score zero on *both* the waypoint and intent dimensions on the majority of reroute samples—21/30 (Opus) and 18/30 (Sonnet)—despite reading the loud half at F1 1.00; their mean reroute F1 over the six-instance extension is low and high-variance (Opus waypoint 0.26±0.25, intent 0.11±0.13; Sonnet 0.30±0.16, 0.20±0.13; n=6 point

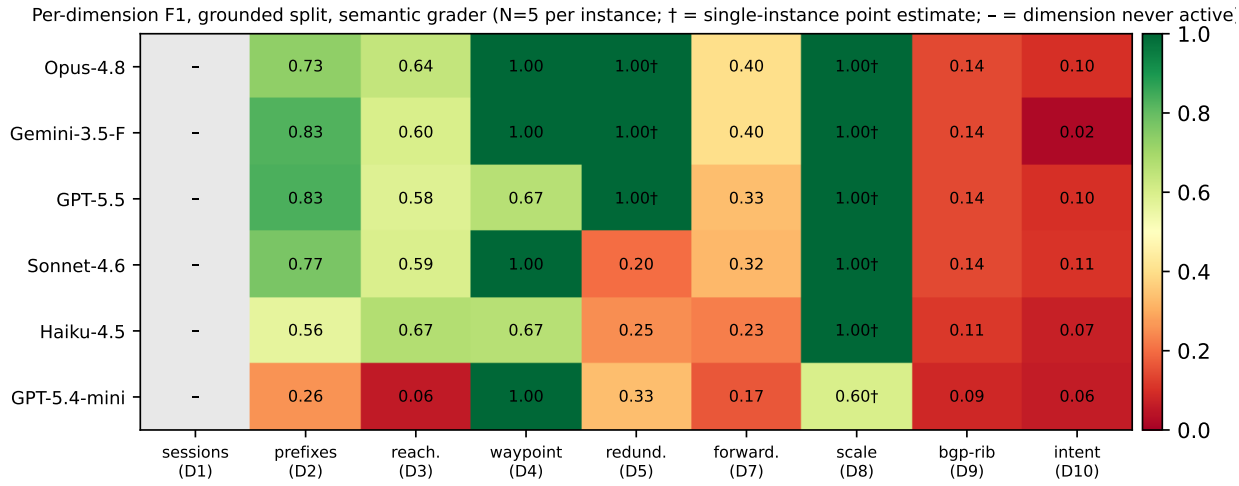


Figure 1: Per-dimension F1 on the grounded split (semantic grader), as *active-convention means*—precision and recall combined over every instance where the dimension is active in truth or prediction. The consequence-dimension cells (D7, D9, D10) are depressed by taxonomy-boundary over-prediction, *not* by missed detection: F1 where each is the ground truth is high (Table 2, F3). The figure is shown to expose that precision gap, not as a detection score.

estimates). Across all four models evaluated on the reroute, 96/120 samples score zero on both dimensions—the one place the frontier's otherwise near-perfect reading drops sharply. GPT-5.5 and Gemini were not run on the reroute extension, so this frontier evidence is Anthropic-only (§8). A second, distinct case is aggregate preservation: when a customer's specific prefixes are dropped but a covering /24 survives (*ipc_aggregate*), models catch the prefix loss (D2) yet miss the traffic-pinning intent the surviving aggregate masks (D10 F1 $\leq$0.20 for five of six models). The low *dimensional* means for the consequence dimensions (active-convention D7 0.17–0.40, D9 0.09–0.14, D10 0.02–0.11; Figure 1) are *not* recall failures: they are dominated by taxonomy-boundary false positives—each model predicts a consequence dimension on roughly three (D7), six (D9), and eight (D10) of the eleven grounded instances where the oracle reserves a neighboring label—the same over-prediction the paper discloses for the D3/D7 boundary (F2). The genuine miss is narrow and consistent: damage is read when it deletes or blackholes something nameable, and missed when every object survives and only the path silently shifts—precisely the class that produced the 2019 leak incidents, where traffic kept flowing, just wrongly.

F4: No home-field advantage. The benchmark was developed and red-teamed primarily with Claude-family models, so grader circularity was a live concern. The cross-vendor frontier lands within noise of itself: Opus 4.8 at 0.51, Gemini 3.5 Flash at 0.49, GPT-5.5 at 0.48—a spread of 0.03 against a per-model standard error of $\approx$0.05 (n=11 instances), so the three are statistically indistinguishable. The Anthropic frontier does not outscore the non-Anthropic frontier; the second Anthropic model (Sonnet 4.6, 0.42) scores *below* both non-Anthropic frontier models, so the Claude-built benchmark does not favor the Claude family.

F5: Naive set grading would have reported universal blindness. Under exact-string matching against the oracle's probe-granularity keys, every model's active-convention forwarding-anomaly F1 is 0.00 (no forwarding-anomaly-truth sample earns credit), and the grounded frontier

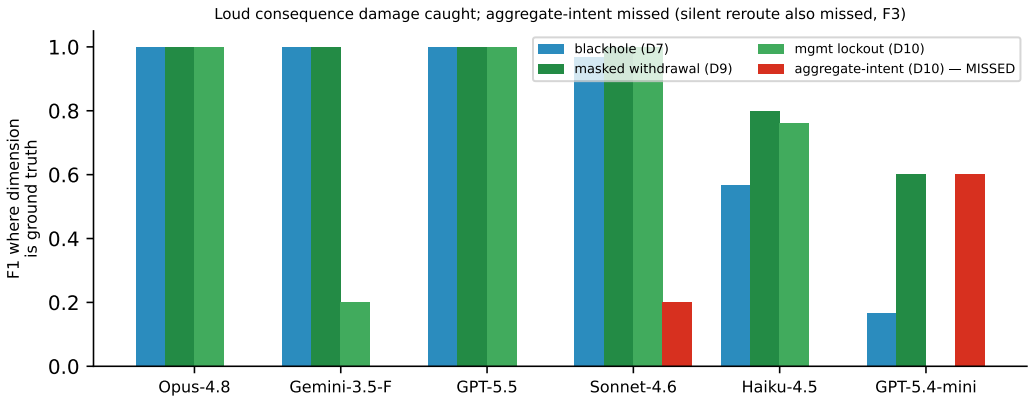


Figure 2: F1 where each consequence dimension is the ground truth (grounded, semantic grader; on these single-truth instances F1 tracks recall). The loud half—blackhole (D7), masked withdrawal (D9, *ipc3_static*), management lockout (D10, *mgmt_lockout*)—is caught by the stronger models; the aggregate-preservation intent (D10, *ipc_aggregate*) is missed. The sharpest quiet case, the leak-induced silent reroute, is missed (F3) and is not plotted here.

macro drops from 0.48–0.51 to 0.36–0.37 (Appendix A)—while the raw outputs contain the correct node, kind, and config-visible prefix. The semantic canonicalization that corrects this is provably the identity on the config-visible dimensions (D1, D2, D8, D9) and leaves every model-free floor at chance (largest movement: the syntactic-diff floor, 0.007→0.023 templated), so the correction lifts models, not floors—and only on the dimensions where the raw outputs were right all along. We report this as a methods finding in its own right: granularity-mismatched string grading is a silent way for a verifier-graded benchmark to manufacture a dramatic, wrong headline.

6 Discussion

Why the quiet half matters most. An assistant that misread routing tables would be caught by any reviewer; an assistant that reads them competently, catches the withdrawal and the lock-out, never false-alarms on sessions (F2), and yet silently misses the leak-induced reroute (F3) is trusted exactly where it is most dangerous—on the one failure mode that leaves no object lost to catch. Binary safety metrics cannot surface this: under a pass/fail rubric, a model that predicts "no impact" on a reroute instance is indistinguishable from one that flags the waypoint bypass, because both made *a* prediction. Typed, graded affected-set scoring is what makes the quiet half measurable—and the quiet half is where the most insidious incidents live: a leak that reroutes a victim's traffic onto a path it should never take, with every session up and every destination still reachable, exactly as in the 2019 route-leak outages.

A candidate graduation test for network agents. As an explicitly novel *position* (a refinement orthogonal to TM Forum's autonomous-network levels), we propose blast-radius prediction as a promotion gate on the autonomy ladder: *advise* → *propose* → *propose-with-self-check* → *gated-execute* → *autonomous*—a system would demonstrate, before advancing past *propose*, that it can enumerate the blast radius of its own changes at a calibrated precision/recall point *across all scored dimensions, including the quiet ones.* No evaluated model meets that bar today: frontier models pass it on the loud half of 4–6 node networks and fail it on the quiet half of the same

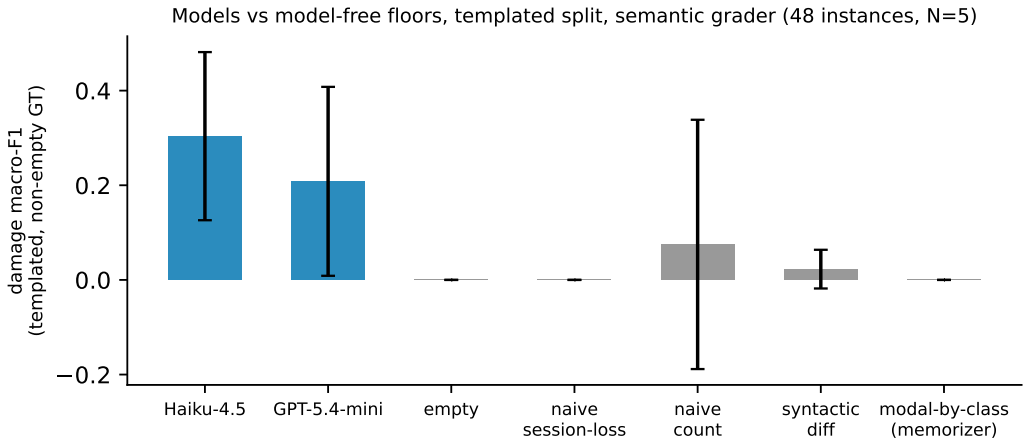


Figure 3: Damage-detection macro-F1 versus model-free floors on the templated split (semantic grader). The modal-by-class memorizer—the strongest template-recall strategy—scores 0.00.

networks. We conjecture the quiet-half gap does not close as topologies grow—the inference chains only lengthen—but scale is untested here.

Open problems. Five follow directly. (1) *Tools-augmented contrast*: how much of the quiet-half gap closes when the model may invoke the verifier, and do agents invoke it on the changes that need it? Cost-aware verifier invocation has been studied in general agentic workflows [13]; its networking instantiation—where a formal oracle defines the invocation-policy optimum—is open, and Asadli et al. point at analyzing agents' tool-usage patterns as future work [1]. (2) *Calibration*: are models *confidently* blind on the silent reroute, or do their confidences flag the gap? (3) *Judge validity*: would an LLM-as-judge grader have caught the granularity artifact of §2, or introduced its own? (4) *Learnability*: parameterized generators plus the oracle yield unlimited perfectly-labeled training pairs; whether supervision closes the quiet-half gap, or the gap survives infinite labels, is an open empirical question. (5) *Contract sharpening*: the D3/D7 boundary (F2) should be stated in the output contract precisely enough that double-typing is unambiguous, without leaking the oracle's disposition taxonomy.

7 Related Work

The rapidly growing use of LLMs across network management and operations is surveyed by Hong et al. [9]; that survey catalogs intent interpretation and task automation but does not treat pre-deployment safety validation—the gap NetBrace targets.

LLM configuration synthesis and repair. NetConfEval translates intents into specifications and code [6]; Mondal et al. study LLM router-configuration synthesis with verifier feedback [7]; PeeringLLM-Bench scores LLM synthesis of vendor-specific BGP peering configurations from policy intent [8]; CORNETTO benchmarks end-to-end configuration repair on 231 instances over real-world topologies of 20–754 nodes [2], and Asadli et al. show agentic architectures with retrieval and Batfish feedback improve repair efficacy (∼12%) and reduce safety regressions (∼17%) over monolithic prompting [1], where the safety metric is a *regression score*: the fraction of the checked

specification set that an *applied* fix newly breaks, computed by the verifier (lower is better). NetBrace measures the complement: whether the *model itself*, with no verifier in the loop, can foresee typed damage from a diff. The verifier is referee, not teammate.

Agentic network benchmarks. NetArena (formerly NetPress) generates dynamic agent tasks and scores correctness, latency, and a rule-adherence action-safety axis in emulation [3]; NetAgentBench formalizes multi-turn configuration tasks as FSMs and measures behavioral stability [4]; NIKA evaluates incident diagnosis on emulated networks [5]; the IRTF NMRG framework draft defines NetConfBench, a 40-task emulator-based suite scored on intent-achievement and functional correctness [10]. Among these, none scores an LLM's *predicted* typed affected set—penalizing both missed and hallucinated damage at the element level—from a static diff, with no verifier in the model's loop. That conjunction is the gap NetBrace fills.

Evaluation methodology. Our design follows the rigor agenda for agentic benchmarks [14]: deterministic non-LLM grading, pinned oracles, contamination controls, explicit floors, and multi-sample reporting. The granularity pitfall of §2 is offered as a concrete addition to that agenda. The holistic multi-metric framing descends from HELM [15].

8 Limitations

Single vendor dialect (FRR) and BGP-centric mechanisms in this release; the packet-filter dimension (D6) is deferred pending a vendor dialect whose ACLs Batfish models as interface filters. The D8 resource *budget* is a declared scenario parameter (the route count is computed; the limit is spec'd, as platform limits are). Topologies are small (4–6 nodes), chosen for oracle determinism; scale is untested, and while we see no mechanism by which the quiet-half reasoning gap would *close* as topologies grow, that expectation is a conjecture, not a result. The grounded split is 11 scenarios; several dimensions appear there once and are reported as point estimates, with the templated split carrying the statistics; the templated split currently has no redundancy-truth (D5) instance, so templated D5 reflects only false-positive behavior. Evaluation is single-shot and tool-free by design; agentic and tools-augmented conditions are the named next step. Batfish models converged steady state, so transient and timer-dependent damage is out of scope. Because grading is deterministic set comparison against a pinned verifier, there is no human-judgment step requiring inter-rater validation; the analogous validity question—does the oracle compute the right ground truth?—is addressed by the determinism gate, the live-FRR spot checks (§2), and the frozen matching specification. A human-*expert baseline* (can a senior operator enumerate the quiet half from the same evidence?) is the named next experiment and is not claimed here. The frontier evidence for the headline silent-reroute miss (F3) is Anthropic-only: the reroute extension ran on Opus 4.8 and Sonnet 4.6, but GPT-5.5 and Gemini were not available for it, so a cross-vendor reroute test—which would also extend the home-field control (F4) to the headline finding—is the first camera-ready item.

9 Conclusion

We introduced typed blast-radius prediction, built an oracle-graded, determinism-gated, judge-free benchmark for it, and measured where current LLMs sit. The line we found is not the one we expected: models—including mid-tier ones—catch the damage that announces itself, up to and including a second-order partial-ECMP blackhole, and never hallucinate a session loss; what they

miss—the one class where even the frontier's near-perfect loud-half reading collapses—is the damage that removes no object and only shifts the path: the leak-induced silent reroute past the inspection node, where the route still resolves, the session stays up, and reachability survives, while they catch the withdrawal, the lockout, and the blackhole, which all delete something nameable. We also showed how easily a verifier-graded benchmark can manufacture the opposite, more dramatic headline through granularity-mismatched string grading, and we release both graders with a frozen matching specification so the correction is inspectable. Until the quiet half closes—or is closed by verifier-in-the-loop architectures whose invocation policies are themselves measured—LLM-proposed network changes need a blast-radius gate that the proposing model cannot currently pass on exactly the failures that matter most. The benchmark, oracle pipeline, run artifacts, and floors make that gate, and progress against it, measurable.

Artifacts. Code, scenario suites, generator seeds, pinned oracle digests, determinism reports, prompts, raw model outputs, graded results under both graders, and the frozen semantic-matching specification will be released alongside the peer-reviewed version; every number in this report traces to a saved run artifact (including the pooled Gemini sample set, both Sonnet runs, and the Opus/Sonnet reroute extension) and will be reproducible from that release.

Acknowledgments and AI disclosure. LLM tooling (Claude) was used extensively under the author's direction for drafting, engineering, and analysis, including adversarial review passes that surfaced the grading artifact reported in §2. All experiments, numbers, and citations were verified by the author, who takes sole responsibility for the content.

A Exact-string grading: the conservative lower bound

Table 3 reports the grounded split under both graders. The exact-string grader demands byte-identical element keys, including the oracle's undisclosed probe-/32 destinations on flow-scoped dimensions; the semantic grader (§2) canonicalizes probes to config-visible covers under a frozen, floor-invariant specification. Dimensions whose truth is already config-visible (D1, D2, D8, D9) are provably identical under both graders, as is the false-session-alarm metric.

	macro-F1 (damage)		forwarding D7 (active conv.)	
Model	exact	semantic	exact	semantic
Claude Opus 4.8	0.37±0.27	0.51±0.17	0.00	0.40
Gemini 3.5 Flash	0.36±0.28	0.49±0.18	0.00	0.40
GPT-5.5	0.36±0.27	0.48±0.17	0.00	0.33
Claude Sonnet 4.6	0.30±0.17	0.42±0.06	0.00	0.32
Claude Haiku 4.5	0.29±0.17	0.39±0.10	0.00	0.23
GPT-5.4-mini	0.22±0.22	0.27±0.21	0.00	0.17

Table 3: Grounded split under both graders. The exact-string D7 column is 0.00 for every model on every forwarding-anomaly-truth sample—the granularity artifact of §2. The semantic D7 column uses the active-instance convention and is depressed below the anomaly-truth recall of Table 2 by the D3/D7 double-typing false positives discussed in F2. Sonnet's grounded rows are from the 12,000-token re-run (Disclosure ii); the exact-string column moves it from 0.30 to 0.42 under semantic matching, the same lift seen fleet-wide.

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